WORKED SOLUTION

STEP II 1999, Question 7

Pure • Curve Sketching

STEP II 1999, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q7

The key idea

Differentiate and put everything over a common denominator: the numerator collapses almost completely: everything quadratic cancels and a single linear factor is left, which settles the stationary points at once. For the sketches, complete the square under the root. That tells us where the curve exists, how it behaves for large $|x|$, and, when $a = 1$, that the square root becomes a modulus, which turns the curve into two pieces of rectangular hyperbola.

Solution

Write $Q(x) = x^2 - 2x + a = (x-1)^2 + (a-1)$.

Exactly one stationary point when $a > 1$

If $a > 1$ then $Q(x) \geq a - 1 > 0$ for every x, so the curve is defined, and differentiable, for all real x. With $y = xQ^{-1/2}$ and $Q'(x) = 2x - 2$, the product rule gives

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= Q^{-1/2} - \tfrac{1}{2}x(2x-2)Q^{-3/2} = \frac{Q - x(x-1)}{Q^{3/2}} \\ &= \frac{x^2 - 2x + a - x^2 + x}{Q^{3/2}} = \frac{a - x}{(x^2 - 2x + a)^{3/2}}.\end{aligned}$$

The denominator is positive and finite for every x, so $\mathrm{d}y/\mathrm{d}x = 0$ exactly when $x = a$. So C has exactly one stationary point, at

$$x = a, \qquad y = \frac{a}{\sqrt{a^2 - a}} = \sqrt{\frac{a}{a-1}},$$

as required. Since $\mathrm{d}y/\mathrm{d}x$ has the sign of $a - x$, the curve rises for $x < a$ and falls for $x > a$, so this point is a maximum.

Ask yourself Where did $a > 1$ get used?

Twice: it makes $Q > 0$ everywhere, so there are no gaps in the curve and no points where the derivative fails to exist; and it makes $x = a$ a genuine point of the curve, since $Q(a) = a^2 - a > 0$. When $a = 1$ the would-be stationary point $x = a = 1$ is exactly where $Q = 0$, and that is what goes wrong in part (ii).

Two more facts help both sketches. For large $|x|$, $\sqrt{Q} = |x|\sqrt{1 - 2/x + a/x^2} \approx |x|$, so

$$y \to 1 \text{ as } x \to +\infty, \qquad y \to -1 \text{ as } x \to -\infty.$$

Also y has the sign of x, and the curve passes through the origin with gradient $a/a^{3/2} = 1/\sqrt{a}$.

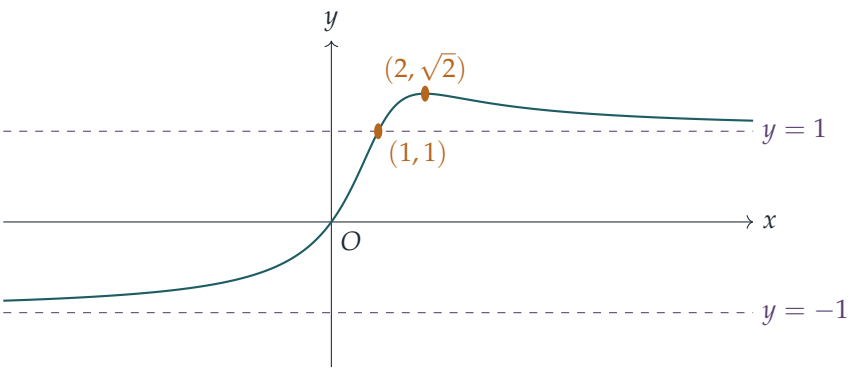
Part (i): $a = 2$

Here $y = \dfrac{x}{\sqrt{(x-1)^2 + 1}}$, defined for all x, with

- a maximum at $(2, \sqrt{2})$, increasing on $x < 2$ and decreasing on $x > 2$;
- the origin on the curve, with gradient $1/\sqrt{2}$ there;
- horizontal asymptotes $y = 1$ (as $x \to \infty$) and $y = -1$ (as $x \to -\infty$).

Which side of each asymptote is the curve on? Setting $y = \pm 1$ and squaring gives $x^2 = x^2 - 2x + 2$, so $x = 1$, and only $y = +1$ fits (since y has the sign of x). So the curve crosses $y = 1$ once, at $(1, 1)$: it is above $y = 1$ for all $x > 1$, coming down towards the asymptote from above after the maximum. It never meets $y = -1$, so it stays above $y = -1$ and rises away from it as x increases from $-\infty$.

(If you want more detail: differentiating again gives $\mathrm{d}^2y/\mathrm{d}x^2 = (2x^2 - 7x + 4)/Q^{5/2}$, so there are points of inflection at $x = \frac{1}{4}(7 \pm \sqrt{17})$, roughly $x \approx 0.72$ and $x \approx 2.78$. The sketch doesn't need these, but they explain its shape.)



Part (ii): $a = 1$

Now $Q = (x-1)^2$, so $x = 1$ is excluded and, because the square root is the positive one,

$$\sqrt{(x-1)^2} = |x-1|.$$

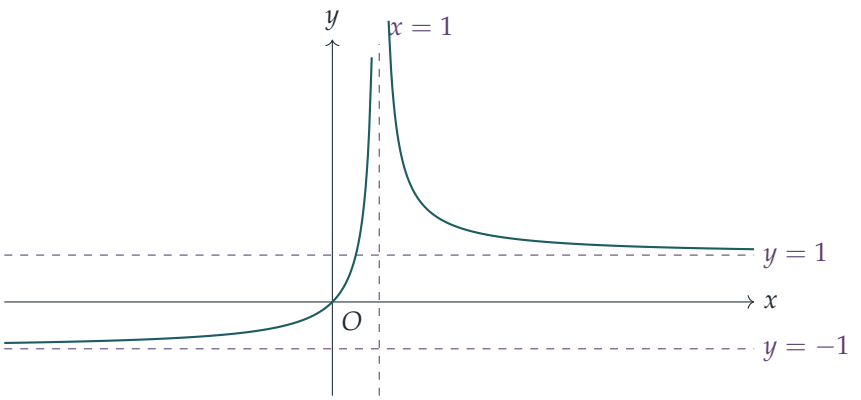
So

$$y = \frac{x}{|x-1|} = \begin{cases} \dfrac{x}{1-x} = -1 + \dfrac{1}{1-x}, & x < 1, \\ \dfrac{x}{x-1} = 1 + \dfrac{1}{x-1}, & x > 1. \end{cases}$$

Each branch is part of a rectangular hyperbola centred at $x = 1$.

- For $x < 1$: y increases (its derivative is $1/(1-x)^2 > 0$), from -1 (approached from above as $x \to -\infty$) through the origin with gradient 1, to $+\infty$ as $x \to 1^-$.
- For $x > 1$: y decreases (derivative $-1/(x-1)^2 < 0$), from $+\infty$ as $x \to 1^+$ down towards 1 (approached from above as $x \to \infty$).

The line $x = 1$ is a vertical asymptote with $y \to +\infty$ on *both* sides (the numerator is near 1, the denominator is small and positive). There is no stationary point: the maximum of the $a > 1$ curves, at height $\sqrt{a/(a-1)}$, has been pushed off to infinity as $a \to 1^+$. The general formula agrees, since $\mathrm{d}y/\mathrm{d}x = (1-x)/|x-1|^3$ is never zero.



Checks

- Numerically, the derivative formula $(a-x)/Q^{3/2}$ agrees with a finite-difference derivative at thousands of random points, and a sign scan finds exactly one sign change for several values of $a > 1$.
- For $a = 2$: $y(2) = 2/\sqrt{2} = \sqrt{2} \approx 1.414$, $y(1) = 1$, and $y(10^6) \approx 1.000001$, $y(-10^6) \approx -0.999999$, confirming the sides from which the asymptotes are approached.
- For $a = 1$: $y(0.999) \approx 999$ and $y(1.001) \approx 1001$, both large and positive, as claimed.

- As $a \to \infty$, the maximum height $\sqrt{a/(a-1)}$ tends to 1, which is sensible: for huge a the curve is nearly $x/\sqrt{a}$ near the origin and the bump flattens.

Common slips

- Writing $\sqrt{(x-1)^2} = x - 1$ in part (ii). That gives $y = x/(x-1)$ everywhere and the wrong left branch, which would then tend to $+1$ as $x \to -\infty$ and go to $-\infty$ as $x \to 1^-$.
- Giving both ends of the curve the asymptote $y = 1$: since $\sqrt{x^2} = |x|$, the limit as $x \to -\infty$ is -1.
- Forgetting to show that the denominator never vanishes when $a > 1$; without that, "the numerator is zero only at $x = a$" does not yet prove there is exactly one stationary point.

Beyond STEP

The family of curves here changes character suddenly at $a = 1$: a smooth curve with a single maximum turns into two separate branches with a vertical asymptote. Studying how the qualitative shape of a family depends on a parameter is the starting point of bifurcation theory, and the care needed with $\sqrt{x^2} = |x|$ and limits as $x \to \pm\infty$ is exactly the care that first-year analysis makes precise. Abbott's *Understanding Analysis* is a friendly place to see limits and continuity done properly.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.